

Security risk assessment report

Part 1: Select up to three hardening tools and methods to implement

In order to prevent future major data breaches and after having evaluated the biggest vulnerabilities inside the organization, the next hardening actions must be implemented to secure the organization network:

The most important ones, with the highest level of priority for implementation, must be:

- the implementation of Multi-Factor Authentication (MFA),
- some hard password policies
- and especially Network Access Privileges.

All of this in order to strengthen and protect the organization's network, leading to hardening security from potential threats, both internally and externally.

Moving on, the priority must lead to these next implementations:

- encryption using the latest standards for data stored,
- Firewall maintenance as a first line of defence from external threats
- and finally port filtering, in order to limit possible penetration from malicious actors.

Part 2: Explain your recommendations

Here is a brief explanation for each recommended implementation to strengthen the organization's network:

MFA: the use of MFA, will significantly reduce the possibility of intrusions from internal employees to other accounts using brute force attacks, and this will reduce password sharing danger, as it will add an extra layer of security. This hardening must be performed usually once.

Password Policies: This will make it more difficult to try brute force attacks, this procedure must be done regularly in order to change the passwords often and

making them complicated. This will be extremely effective in order to protect the database, as the admin password is still set to default.

Network access privileges: This will be the most effective way to protect the internal database and avoid possible internal brute force attacks, all of this by permitting, limiting, and/or blocking access privileges to network assets for people and roles, so that the sensitive part of the network will remain safe and protected. This can be done once and changed accordingly to possible social engineering attacks.

Encryption using the latest standards: This must be done as an extra layer of defences in order to be more difficult to read sensitive data, moving in and out from the organization's network, and if a breach occurred. This must be updated regularly.

Firewall maintenance: This will be the first layer of defences in order to filter data traffic coming in and out of the organization's network, limiting the possibilities of external intrusions and threats. This is updated regularly in order to face threats in the future and new ones.

Port Filtering: This is a layer of extra defence in order to avoid possible intrusions using unused ports as the entrance to the network. This will be updated accordingly to the needs of the organization and the firewall updates.